

Lesson Plan — Electrostatics (Electric charge, Coulomb's law, electric field, Gauss's law)

Physics | Grade 11-12 | Foundational | Electric Charge and Field

Overview

Subject	Physics
Grade band	11-12
Topic	Electrostatics (Electric charge, Coulomb's law, electric field, Gauss's law)
Chapter	Electric Charge and Field
Source document	1.4.1 Additivity of charges
Periods	6 x 40 min

Learning Objectives

- [apply] Calculate the magnitude and direction of the electrostatic force between two point charges using Coulomb's law for given q_1 , q_2 and separation r .
- [analyze] Derive the electric field at a point due to a continuous charge distribution by integrating Coulomb's law over the distribution (expressed using ρ , σ or λ as appropriate).
- [apply] Use Gauss's law to compute the electric field for high-symmetry charge distributions: infinite plane, infinite line, and thin spherical shell, stating and justifying the Gaussian surface chosen.
- [understand] Predict and compute the torque on a permanent electric dipole in a uniform electric field and decide whether a net force exists for uniform versus non-uniform fields.
- [apply] Compute numerical charge amounts using the quantisation relation $q = n e$ and convert between elementary charge units and coulombs for simple problems (including order-of-magnitude estimates).
- [apply] Apply the superposition principle to find the net force or net electric field at a point due to multiple point charges by vector addition.

Period-by-Period Plan

Period 1: Quantised Electric Charge and Units

Concepts	Electrostatics, Electric charge, Quantisation of charge
Objectives	Compute numerical charge amounts using the quantisation relation $q = n e$ and convert between elementary charge units and coulombs for simple problems (including order-of-magnitude estimates).
Time allocation	Activate prior ideas (5 min), Demonstration and measurement (8 min), Derivation (10 min), Guided practice (8 min), Formative check (6 min),

Why this order	Consolidation and homework (3 min) As the opening lesson, this period establishes the microscopic idea of electric charge and the quantisation relation $q = n e$ (where q is in coulombs, $e \sim 1.60 \times 10^{-19} \text{ C}$), which is needed before studying Conductors and insulators, the Principle of superposition, and defining the Electric field. Mastery of converting between numbers of elementary charges and coulombs makes numerical work and order-of-magnitude estimates in later lessons possible.
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Period 2: Conductors, Superposition and Electric Field

Concepts	Conductors and insulators, Principle of superposition, Electric field
Objectives	Compute numerical charge amounts using the quantisation relation $q = n e$ and convert between elementary charge units and coulombs for simple problems (including order-of-magnitude estimates).
Time allocation	Activate prior knowledge (5 min), Introduce new concepts (12 min), Introduce superposition and field (10 min), Guided practice (8 min), Check for understanding (3 min), Consolidate and connect forward (2 min)
Why this order	This period builds directly on Quantised Electric Charge and Units by applying quantisation to multiple-charge setups and introducing how charges behave in conductors versus insulators; it prepares students to treat source charge and test charge, apply the superposition principle to electric fields, and compute flux for dipole-like arrangements.

Period 3: Superposition, Flux and Electric Dipole

Concepts	Source charge and test charge, Superposition principle for electric fields, Electric flux through a surface, Electric dipole
Objectives	Compute numerical charge amounts using the quantisation relation $q = n e$ and convert between elementary charge units and coulombs for simple problems (including order-of-magnitude estimates).
Time allocation	Activate prior knowledge (6 min), Introduce electric flux (12 min), Introduce electric dipole (10 min), Guided practice / demonstration (6 min), Check and consolidate (6 min)
Why this order	This period builds on the prior lesson "Conductors, Superposition and Electric Field" by using the idea of a fixed source charge and vector superposition to compute net fields and by introducing electric flux as $E \cdot dS$. Mastery of dipole definition and flux here makes it possible to compare dipole and monopole field fall-off and to apply Gauss's law and Coulomb's law in the following lesson.

Period 4: Dipole Field, Gauss's Law & Charges

Concepts	Dipole field falls off faster than monopole, Gauss's law, Coulomb's law
Objectives	Compute numerical charge amounts using the quantisation relation $q = n e$ and convert between elementary charge units and coulombs for

Time allocation	simple problems (including order-of-magnitude estimates). Activate prior knowledge (5 min), Introduce new derivation (10 min), Demonstration and measurement (7 min), Guided practice (10 min), Quick check (formative) (5 min), Consolidation (3 min)
Why this order	This period builds on the prior work on superposition, electric flux and the basic electric dipole to derive how dipole fields scale and to apply Coulomb's law in quantitative examples. Mastery of dipole field falloff and numerical charge calculations enables the next material on continuous charge distributions, field lines and torque on a dipole in a uniform field.

Period 5: Electric Field, Lines, Flux & Dipoles

Concepts	Electric field, Electric field lines, Electric flux, Continuous charge distributions, Torque on a dipole in a uniform electric field
Objectives	Calculate the magnitude and direction of the electrostatic force between two point charges using Coulomb's law for given q_1 , q_2 and separation r .; Predict and compute the torque on a permanent electric dipole in a uniform electric field and decide whether a net force exists for uniform versus non-uniform fields.; Apply the superposition principle to find the net force or net electric field at a point due to multiple point charges by vector addition.
Time allocation	Activate prior knowledge (5 min), Introduce electric field and field lines (12 min), Demonstration/mini-experiment (8 min), Guided practice / problem set (10 min), Check and consolidate (5 min)
Why this order	This period builds on the previously developed Coulomb's law, dipole versus monopole field falloff, and Gauss's law (Dipole Field, Gauss's Law & Charges) by introducing the field concept, field lines, flux and continuous charge densities so students can set up and compute fields by superposition and integration in the next period (Field of an infinitely long uniformly charged wire; plane sheet; spherical shell; continuous distributions).

Period 6: Fields of Line, Plane and Shell

Concepts	Field of an infinitely long uniformly charged wire, Field of a uniformly charged infinite plane sheet, Field of a uniformly charged thin spherical shell, Field from continuous distribution via superposition/integration
Objectives	Derive the electric field at a point due to a continuous charge distribution by integrating Coulomb's law over the distribution (expressed using ρ , σ or λ as appropriate).; Use Gauss's law to compute the electric field for high-symmetry charge distributions: infinite plane, infinite line, and thin spherical shell, stating and justifying the Gaussian surface chosen.
Time allocation	Activate prior knowledge (5 min), Introduce new derivation (10 min), Introduce complementary derivations (15 min), Practice by integration

Why this order

and superposition (5 min), Check understanding and consolidate (5 min)
Builds directly on 'Electric Field, Lines, Flux & Dipoles' by using Coulomb's law and flux concepts to derive fields for symmetric continuous distributions; this enables applying Gauss's law and direct integration to compute the field of an infinite line, infinite plane, and thin spherical shell as closing applications.

Concepts Taught

Concept	Importance	Summary
Electrostatics	core	Study of forces, fields and potentials arising from static charges.
Electric charge	core	An attribute of matter responsible for electric interactions; comes in two polarities and can be transferred between bodies by rubbing.
Quantisation of charge	core	Observable free charges occur in integer multiples of a basic unit e (magnitude of electron/proton charge).
Conductors and insulators	core	Materials classified by whether they allow electric charges (electrons) to move freely (conductors) or not (insulators); charges on conductors redistribute over the surface while on insulators they stay localized.

Concept	Importance	Summary
Principle of superposition	core	Net electrostatic force on a charge from multiple other charges equals the vector sum of the forces due to each charge taken individually; individual pairwise forces are unaffected by presence of others.
Electric field	supporting	A vector field produced by a charge that exerts a force on other charges; the field at a position r due to a point charge Q at the origin is specified by a position-dependent vector $E(r)$.
Electric field	core	Electric field is a vector field in space that characterises the electrical environment of a system of charges; at a point it is defined operationally as the force per unit positive test charge placed there (without disturbing the system).
Source charge and test charge	supporting	A source charge produces the electric field; a test charge probes the field. The source charge must be considered fixed (or negligibly disturbed) when defining the field produced.

Concept	Importance	Summary
Superposition principle for electric fields	core	The net electric field due to a system of point charges at a point is the vector sum of the fields due to the individual charges evaluated at that point.
Electric field lines	core	Field lines are curves whose tangent at any point gives the direction of the electric field there; density of lines represents relative field strength and they obey specific properties (start on + charges, end on - charges, do not cross, etc.).
Electric flux through a surface	supporting	Electric flux through a surface element is the scalar product of the electric field and the area vector: it is proportional to the number of field lines cutting the element and equals $\mathbf{E} \cdot d\mathbf{S} = E dS \cos \theta$ for an element.
Electric dipole	core	An electric dipole is a pair of equal and opposite point charges $\pm q$ separated by distance $2a$; the dipole moment vector points from the negative to the positive charge and has magnitude $q \times 2a$.

Concept	Importance	Summary
Dipole field falls off faster than monopole	core	At distances r much larger than the dipole separation ($r \gg a$), the dipole electric field falls off as $1/r^3$ (not $1/r^2$) and depends on the dipole moment $p = q(2a)$.
Gauss's law	core	Electric flux through a closed surface equals the total enclosed charge divided by ϵ_0 .
Coulomb's law	core	Mutual electrostatic force between two point charges is proportional to the product of charges and inversely proportional to the square of their separation; constant $k = 1/(4\pi\epsilon_0)$.
Electric flux	supporting	Flux through a small area element dS equals $\mathbf{E} \cdot d\mathbf{S}$, where vector area element $d\mathbf{S}$ is $\hat{n}$ times area magnitude.
Continuous charge distributions	charge core	Macroscopic charge distributions are described by linear (λ), surface (σ), and volume (ρ) charge densities defined as charge per chosen macroscopic element.
Torque on a dipole in a uniform electric field	supporting	A permanent electric dipole with dipole moment p in uniform field E experiences torque $\tau = p \times E$ and no net force.

Concept	Importance	Summary
Field of an infinitely long uniformly charged wire	core	Electric field at radial distance r from an infinite line charge of linear density λ is $E = (\lambda)/(2 \pi \epsilon_0 r)$ in the radial direction.
Field of a uniformly charged infinite plane sheet	core	Electric field due to an infinite plane sheet with uniform surface charge density σ has magnitude $E = \sigma/(2 \epsilon_0)$ directed normal to the plane.
Field of a uniformly charged thin spherical shell	core	Outside the shell ($r > R$) the field is as if the total charge q is concentrated at the centre: $E = q/(4 \pi \epsilon_0 r^2) \hat{r}$. Inside the shell ($r < R$) the field is zero.
Field from continuous distribution via superposition/integration	supporting	Electric field from a continuous charge distribution is obtained by summing (integrating) contributions dE from volume elements ρdV using Coulomb's law.

Formulae

Quantisation of charge

$$q = n e$$

where q = charge on a body (C); n = integer (positive, negative or zero); e = elementary charge (magnitude of proton/electron charge) (C)

Coulomb's law (magnitude)

$$F = k (q_1 q_2) / r^2 = (1 / 4\pi\epsilon_0) (q_1 q_2) / r^2$$

where F = magnitude of electrostatic force between two point charges (N); q_1, q_2 = point charges (C); r = distance between the charges (m); k = Coulomb constant ($1/4\pi\epsilon_0$) ($N \cdot m^2 \cdot C^{-2}$); ϵ_0 = permittivity of free space ($C^2 \cdot N^{-1} \cdot m^{-2}$)

Electric field of a point charge

$$E(r) = (1 / 4\pi\epsilon_0) (Q \hat{r} / r^2)$$

where $E(r)$ = electric field vector at position r ($N \cdot C^{-1}$ (or $V \cdot m^{-1}$)); Q = source point charge located at the origin (C); r = position vector from origin to field point (magnitude r) (m); $\hat{r}$ = unit vector from origin to field point; ϵ_0 = permittivity of free space ($C^2 \cdot N^{-1} \cdot m^{-2}$)

Force on a charge in an electric field

$$F(r) = q E(r)$$

where F = Force on the test charge (N); q = Test charge (C); E = Electric field at position r ($N \cdot C^{-1}$)

Electric flux through an area element

$$d\Phi = E \cdot dS = E dS \cos \theta$$

where Φ = Electric flux ($N \cdot C \cdot m$); E = Electric field ($N \cdot C^{-1}$); dS = Area element vector (magnitude dS , direction normal) (m^2); θ = Angle between E and area normal (rad)

Dipole moment

$$p = q \times 2a \text{ (vector from } -q \text{ to } +q)$$

where p = Dipole moment vector (C m); q = Magnitude of each charge (C); $2a$ = Separation between charges (m)

Coulomb force between two point charges

$$F = (1 / (4 \pi \epsilon_0)) * (q_1 q_2 / r^2) \hat{r}$$

where q_1, q_2 = point charges (C); r = distance between charges (m); ϵ_0 = permittivity of free space ($F \cdot m^{-1} (C^2 \cdot N^{-1} \cdot m^{-2})$)

Gauss's law (integral form)

$$\text{Flux through closed surface} = q_{\text{enclosed}} / \epsilon_0$$

where Φ_E = electric flux through closed surface S ($N \cdot m^2 \cdot C^{-1}$); q_{enc} = total charge enclosed by S (C)

Torque on a dipole in uniform field

$$\tau = p \text{ cross } E$$

where p = electric dipole moment (C m); E = electric field ($N \cdot C^{-1}$)

Field of infinite line charge

$$E = \lambda / (2 \pi \epsilon_0 r) \text{ radial unit vector}$$

where λ = linear charge density ($C \cdot m^{-1}$); r = radial distance from wire (m)

Field of infinite plane sheet

$$E = \sigma / (2 \epsilon_0) \text{ normal unit vector}$$

where σ = surface charge density ($C \cdot m^{-2}$)

Field of uniformly charged thin spherical shell (outside)

$$E = q / (4 \pi \epsilon_0 r^2) \hat{r} \text{ for } r > R$$

where q = total charge on shell (C); r = distance from centre (m)

Field inside thin spherical shell

$$E = 0 \text{ for } r < R$$

where r = distance from centre (m); R = shell radius (m)